

Husnain Nadeem

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EDUCATION

National Textile University

Bachelor of Science in Computer Science

Faisalabad, PK

Oct 2023 – July 2027

- Undergraduate Coursework: Data Structures; Object-Oriented Programming; Information Security; Artificial Intelligence; Web Engineering; IoT Systems; Database Systems; Discrete Mathematics.

UNDERGRADUATE PROJECTS

HMH Game — Semester 1

2023

C++ · Game Development

github.com/Husnain-Nadeem

- Collaborated with team to build a fully functional desktop game in C++ from scratch covering core game loop, rendering, and player mechanics.
- Practised object-oriented design and team-based version control across multiple development cycles.

EasyKhatta Replica — Semester 2

2024

C# · MySQL · Desktop Application

- Built a replica of the EasyKhatta accounting application using C# with a MySQL backend.
- Implemented full CRUD operations, user authentication, and ledger management features.

Alarm Capsule — Semester 3

2024

C++ · Data Structures · Doubly Linked List

- Developed an alarm management system using a doubly linked list as the core data structure.
- Focused on time complexity and efficient insertion, deletion, and traversal of alarm nodes.

Network Traffic Analyser — Semester 4

2024

Information Security · Python · Networking

- Built a basic network packet analyser similar to Wireshark that monitors inbound and outbound traffic.
- Captured and displayed real-time packet data to support understanding of network security fundamentals.

AI Video Generation System — Semester 4

2024

Python · HuggingFace · Gemini API · AI/ML

- Downloaded and integrated a HuggingFace video generation model, enabling video creation from natural language commands.
- Attached the Gemini API to parse and refine user instructions before passing them to the model, improving output accuracy.

Smart Home IoT System — Semester 5

2024

IoT · IFTTT · Google Assistant · Arduino · Sensors

- Built an end-to-end smart home system integrating lights, temperature control, and sonar sensors.
- Connected Google Assistant via IFTTT for voice-controlled automation of all home devices.

Automated Accounting System — Semester 6

2025

Python · Flask · HTML · CSS · JavaScript · MySQL

- Developed a full-stack web application where users enter financial data and the system auto-generates the complete accounting cycle.
- Implemented automated journal entries, ledger, trial balance, and Excel export functionality.

CLOUD AND DEVOPS PROJECTS

End-to-End CI/CD Pipeline — Jenkins, ArgoCD, Kubernetes

2026

Jenkins · Maven · SonarQube · Docker · ArgoCD · Minikube · EKS

- Built a full GitOps pipeline from code commit to Kubernetes deployment — build, test, Docker image publish, and ArgoCD sync. Debugged 12 pipeline failures before successful run.
- Deployed a game application on AWS EKS with Fargate, ALB Ingress Controller installed via Helm, and IAM OIDC for pod-level AWS permissions.

AWS Native CI/CD — CodePipeline, CodeBuild, CodeDeploy

2026

AWS CodePipeline · CodeBuild · CodeDeploy · EC2 · IAM Roles

- Built a fully AWS-native CI/CD pipeline: GitHub as source, CodePipeline as orchestrator, CodeBuild for build and scan, CodeDeploy to EC2 agent.
- Configured IAM roles for every service-to-service communication link — the primary debugging challenge across the project.

AWS Production Architecture — Multi-AZ VPC, ALB, Auto Scaling

2026

AWS · EC2 · VPC · ELB · Route 53 · Bastion Host

- Multi-AZ VPC with ALB, Auto Scaling (1-4 instances), NAT Gateways, Bastion Host SSH jump access, and Python HTTP server load-balanced across private EC2 instances.

CloudWatch Cost Optimisation — Lambda + boto3

2026

AWS CloudWatch · Lambda · Python · boto3

- Serverless system triggered by CloudWatch daily: Lambda scans for unattached EBS volumes and orphaned snapshots and deletes them automatically, reducing unnecessary AWS spend.

AWS Resource Checker Script

2025

Bash · AWS CLI · Linux

- Bash script that checks status of EC2, S3, IAM, and VPC resources across an AWS account and outputs a readable report. github.com/Husnain-Nadeem/Aws-resource-checker-script

CERTIFICATIONS

AWS Cloud Practitioner Essentials

March 2025

Amazon Web Services

Job Roles in the Cloud

February 2025

Amazon Web Services

CO-CURRICULAR ACTIVITIES

External Affairs — Computer Science Society

2024 - Present

National Textile University

- Manage external communications, outreach, and partnerships on behalf of the CS Society.

DCS Ambassador — Quran Society

2023 - Present

National Textile University

- Active member contributing to society events and community engagement on campus.

TECHNICAL SKILLS

Cloud: AWS — EC2, VPC, IAM, S3, ELB, Route 53, Auto Scaling, NAT Gateway, CloudFormation, CodePipeline, CodeBuild, CodeDeploy, CloudWatch, Lambda

CI/CD & DevOps: Jenkins · ArgoCD · GitOps · Groovy · SonarQube · Maven · Docker · Docker Hub · Minikube · kubectl · Helm · eksctl

Kubernetes & EKS: Pod management · Deployments · Services · Ingress · Fargate · ALB Controller · IAM OIDC

IaC: AWS CloudFormation (YAML/JSON) · Terraform (learning) · Ansible (learning)

Linux & Scripting: Linux administration · Bash scripting · SSH · AWS CLI · boto3

Web Development: Python Flask · HTML · CSS · JavaScript · MySQL

Programming: Python · C++ · C# · C · Bash

Version Control: Git · GitHub